

Final Project

This exam is open-book but must be completed independently. You may not discuss the problems with anyone other than the instructor (lmao@biostat.wisc.edu).

For this project, you will be analyzing a dataset from a randomized, controlled trial among women of childbearing age to evaluate the effects of an educational intervention. One response variable of interest is the participants' self-rating of health as either good or poor. The researchers would like to assess the effect of the intervention on self-rated health across the follow-up period, as well as whether these effects are influenced by the mother's age. The file `HEALTH.txt` contains data on $n=80$ women enrolled in this trial. These data were measured at 4 points in time: randomization, 3 months, 6 months, and 12 months post-randomization. Each observation in the dataset contains values for the following variables (in this order):

- ID: unique participant identification code
- TIME: the visit number for this observation of this participant
 - 1 corresponds to a participant's visit at the time of randomization
 - 2 corresponds to the 3 month post-randomization visit
 - 3 corresponds to the 6 month post-randomization visit
 - 4 corresponds to the 12 month post-randomization visit
- RX: the group to which the participant has been randomized
 - Control
 - Intervention
- HEALTH: participants self-rated level of health for this visit
 - Good
 - Poor
- AGEGROUP: participants age group at time of randomization
 - 15-24 (years old)
 - 25 to 34 (years old)
 - 35+ (years old)

Write a data-analysis report by following the steps below (and append your annotated code).

- (a) Tabulate descriptive statistics for all relevant variables. Do this for the overall sample, as well as by intervention group. Provide a graphical representation of longitudinal trend in self-rated level of health by intervention group.
- (b) Fit a repeated measures model across all study follow-up visits (**but not at randomization**) to describe the marginal relationship of the participants' self-ratings as a function of the effects of randomized group, visit, **health self-rating at randomization**, and age group as explanatory variables. Consider any pairwise interactions among these variables for inclusion in the final model. Determine your best estimate for the intervention effect at each follow-up visit. Present and justify your choice for the final model, incorporating a discussion of any assumptions or limitations of the model.

(Hint: You may want to revisit slides of Lecture 6 for how to analyze longitudinal data adjusting for baseline response.)

- (c) [Extra credit] Investigators are concerned that subjects may be less likely to continue in the study if they have poor health self-ratings. Investigate the pattern of missing data. As sensitivity analysis, re-analyze the data by multiple imputations or inverse probability weighting to adjust for potentially informative dropout using prior information. Compare the results with those from (b).
- (d) Using the most appropriate model determined in (b), provide one to two concise paragraphs suitable for the Methods section of a manuscript that clearly describe the models fit to the data. In addition, provide two to three paragraphs (plus one to three related tables or figures) for the manuscript's Results section that give a comprehensive summary of your findings, including the interpretation of noteworthy odds ratio estimates (and corresponding 95% confidence intervals). For significant interaction effects, if any, be sure to fully interpret the nature of those interactions.
- (e) Repeat your summary of the most important findings of the study, but this time in the form of a press release for the mass media. Be certain to use concise, nontechnical language that would be understandable to a lay reader, while still accurate scientifically. Also, be sure to address whether the effects of the intervention vary across time or by age group.